

Jake Doherty
(954) 829-1707 | jakedoherty@utexas.edu

Education

BS , Physics, University of South Florida, Tampa, FL, US	May 2022
BA , Mathematics, University of South Florida, Tampa, FL, US	May 2022
PhD , Physics, University of Texas at Austin, Austin, TX, US	2022–Present

Training

Research Trainee , Department of Physics, USF Optical and Spectroscopy Laboratory, University of South Florida, Tampa, FL Advisors: Denis Karauskaj, PhD; Varun Mapara	09/2019–12/2019
Research Assistant , Department of Physics, University of South Florida, Tampa, FL Advisor: Inna Ponomareva, PhD	05/2020–08/2022
Graduate Researcher , Department of Physics, University of Texas at Austin, Austin, TX Advisor: Pablo Laguna, PhD	08/2022–Present

Publications

Tunability of Structure, Polarization, and Band Gap of High T_c Organic-Inorganic Ferroelectrics by Hydrostatic Pressure

P. S. Ghosh, **J. Doherty**, S. Lisenkov, and I. Ponomareva
The Journal of Physical Chemistry C 2021 125 (29), 16296-16303
DOI: 10.1021/acs.jpcc.1c03980

Negative Capacitance Regime in Antiferroelectric PbZrO_3
J. Appl. Phys. 132, 034101 (2022); <https://doi.org/10.1063/5.0087290>
J. Doherty, K.A. Lynch, and I. Ponomareva

Growth of a Black Hole in a Scalar Field Cosmology
Jake Doherty (Texas U.), Miguel Gracia-Linares (Texas U.), Pablo Laguna (Texas U.)
Phys. Rev. D 112, 044029 (2025)
DOI: <https://doi.org/10.1103/nypy-cd2f>

Talks at Conferences

American Physical Society Global Physics Summit, Denver, CL	2026
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Honors and Awards

USF Dean's List of Scholars	2021
USF Director's Award	2018
USF FL Academic Scholars Award	2018

Teaching and Advising Experience

Teaching Assistant, Department of Physics, University of South Florida	01/2021–05/2022
Teaching Assistant, Department of Physics, University of Texas at Austin	08/2022–Present
Directed Reading Program mentor, Department of Physics, University of Texas at Austin	08/2022–Present

Selected Coursework

Linear Algebra, Mathematical Methods of Physics, Fundamental techniques in Machine Learning and Data Science

Technical Skills

Programming: Advanced: Python; Intermediate: C/C++, Fortran, Bash

Scientific Computing: UN*X, Slurm, NumPy, matplotlib

Research: Numerical simulations, theoretical modeling of physical systems